# scholia

## Educational Document Formatting for Mathematics Pedagogy

Esther Poniatowski
esther.poniatowski@ens.psl.eu

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**Abstract**

SCHOLIA is a LaTeX package for documents requiring consistent visual structure. It provides callout boxes and inline frames for highlighting content, and numbering systems to organize steps and components. For educational use, it adds exercise environments, grading tools, and worksheet layouts. The package supports article and Beamer modes with French/English localization.

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# I. Overview

SCHOLIA provides environments and commands for documents requiring structured visual components. Typical applications include educational materials, technical tutorials, reports, and academic projects.

## 1. Key Features

The package offers six main feature categories, listed from most general (applicable to any structured document) to most specific (educational content):

| Feature | Description |
|---|---|
| Callouts | 13 styled box types (definitions, theorems, warnings, tips, etc.) |
| Reference frames | Inline styled boxes for cross-references |
| Numbering | Steps/substeps and questions/subquestions with hanging indentation |
| Fiches | Standalone worksheet layouts with banners and relative pagination |
| Exercises | 5 environment types with difficulty indicators and solution control |
| Grading | Rubric tables and point allocation for assessments |

## 2. Document Modes

The package is designed to maintain consistent styling across multiple formats. This allows presenting the same content in both printed documents and slide presentations.

Two modes are supported:

| Mode | Purpose | Features |
|---|---|---|
| **Article** (default) | Printed materials, handouts, worksheets | All features enabled. |
| **Beamer** | Presentations, oral sessions | Fiches and reference frames disabled (page-based constructs incompatible with slides). Progression module loaded for slide navigation. |

## 3. Pre-Defined Styles and Customization

SCHOLIA offers two levels of usage:

- **Ready-to-use environments** — Variants in each feature, with pre-defined colors, icons, and styling. No configuration required. (13 callout types, 8 reference frame, 5 exercise categories, 10 fiche layouts)
- **Customization framework** — Generic commands allow creating new variants in each feature. All visual

elements can be redefined (colors, icons, spacing, borders) using the same underlying infrastructure.

# II. Installation

## 1. CTAN Installation

Once available on CTAN, install via the standard TeX distribution tools:

```
# TeX Live
tlmgr install scholia

# MiKTeX
miktex-console --install=scholia
```

## 2. Manual Installation

Place the `scholia/` directory in a local texmf tree:

```
~/texmf/tex/latex/scholia/
```

Then refresh the filename database:

```
texhash ~/texmf
```

# III. Loading the Package

## 1. Basic Usage

SCHOLIA is loaded in the preamble with `\usepackage`. The package detects the document class automatically: with `article`, `report`, or `book`, it operates in article mode; with `beamer`, it switches to presentation mode.

**Standard article with French strings (default):**

```latex
\documentclass{article}
\usepackage{scholia}
```

**Article with English strings:**

```latex
\documentclass{article}
\usepackage[english]{scholia}
```

**Beamer presentation:**

The `beamer` option can be specified explicitly, though it is auto-detected when using the `beamer` document class.

```latex
\documentclass{beamer}
\usepackage[beamer]{scholia}
```

## 2. Package Options

| Option | Description |
|---|---|
| `french` | Use French strings (default) |
| `english` | Use English strings |
| `beamer` | Enable Beamer mode (auto-detected for `beamer` class) |
| `noicons` | Disable icon loading for text-only output |
| `minimal` | Minimal dependencies; implies `noicons` |
| `draft` | Draft mode (reserved for future use) |

## 3. Localization

All user-facing strings (labels, titles, prompts) are defined in localization files. The `french` and `english` options select the appropriate string set.

| Element | French | English |
|---|---|---|
| Exercise label | Exercice | Exercise |
| Definition | Définition | Definition |
| Step | Étape | Step |
| Total | Total | Total |

To override specific strings, redefine the internal macros after loading SCHOLIA:

```latex
\usepackage{scholia}
\def\scholia@str@exo@label{Problem}
```

## 4. Icons

Callouts and exercises display icons by default, loaded from PDF files in the `icons/pdf/` directory. The `noicons` option disables icon loading, which is useful for:

- Faster compilation during drafting
- Reduced output file size
- Plain-text fallback scenarios

# IV. Quick Start

The following example demonstrates SCHOLIA in an article document with a definition callout and a guided exercise. The exercise uses numbered steps and a conditional answer blank.

```latex
\documentclass{article}
\usepackage[english]{scholia}

\begin{document}

\begin{definitionbox}[Derivative]
The derivative of $f$ at $x$ is the limit
$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$
when this limit exists.
\end{definitionbox}

\begin{exerciseguided}[Computing Derivatives]
\exerciseStep{Find $f'(x)$ for $f(x) = x^2$.}

Apply the definition: $\lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$.

The answer is \solutionBlank{2cm}{$2x$}.
\end{exerciseguided}

\end{document}
```

# V. Features

This section documents each feature category in detail.

## 1. Callouts

Callouts are styled boxes for highlighting specific content types. Each callout type provides two environment variants:

- `<type>box` — Full box with optional title and body content
- `<type>title` — Title-only bar (no body area)

### a. Available Types

| Environment | Color | Typical Use |
|---|---|---|
| `definitionbox` | Cyan | Mathematical definitions |
| `theorembox` | Red | Theorems and propositions |
| `proofbox` | Gray | Mathematical proofs |
| `warningbox` | Orange | Warnings, common mistakes |
| `errorbox` | Dark red | What NOT to do |
| `tipbox` | Yellow | Tips and strategies |
| `methodbox` | Blue | Step-by-step methods |
| `notationbox` | Brown-gray | Notation conventions |
| `objectivebox` | Green | Learning objectives |
| `summarybox` | Light gray | Summaries and recaps |
| `remarkbox` | Purple | Observations |
| `questionbox` | Blue | Questions for reflection |
| `instructionbox` | Gray | Instructions and directives |

### b. Usage Examples

**Box with Titl**

```
\begin{definitionbox}[Some title]
Some content here.
\end{definitionbox}
```

**Some title**

Some content here.

**Box without Titl**

```
\begin{tipbox}
Some content here.
\end{tipbox}
```

Some content here.

**Title-Only Varian**

```
\begin{theoremtitle}{Some text}
\end{theoremtitle}
```

**Some text**

The title-only variant accepts an optional first argument for tcolorbox options:

```
\begin{theoremtitle}[centered]{Centered text}
```

```
\end{theoremtitle}
```

### c. Customization

Each callout type is defined by a color. To create a new callout type:

1. Define a color with `\definecolor`
2. Create the environment using `callout layout` (for boxes) or `callout titleonly` (for title-only variants)

```
\definecolor{callout-custom}{RGB}{100,150,200}
\newtcolorbox{custombox}[1][]{%
  callout layout=callout-custom,
  title={#1},
}
```

The styles are based on tcolorbox; any tcolorbox option can be added to further customize spacing, borders, or backgrounds.

## 2. Reference Frames

Reference frames are inline styled boxes for cross-referencing other documents or sections. They provide visual consistency when referring to fiches, methods, or other resources. Available only in article mode.

### a. Generic Frame

The `\inlineframebox` command creates a colored inline frame:

```
See the \inlineframebox[cyan]{theory section} for details.
```

With navigation arrow (starred variant):

```
Continue to \inlineframebox[green]*{exercises}.
```

### b. Typed Reference Frames

Pre-defined commands match fiche types:

| Command | Color/Purpose |
|---|---|
| `\notionsframe{text}` | Cyan (theory references) |
| `\exercicesframe{text}` | Orange (exercise references) |
| `\correctionsframe{text}` | Green (solution references) |
| `\methodeframe{text}` | Blue (method references) |
| `\evaluationframe{text}` | Red (assessment references) |
| `\rechercheframe{text}` | Reddish-orange (research activities) |
| `\autoevaluationframe{text}` | Purple (self-assessment) |
| `\aretenirframe{text}` | Gold (key points) |

All typed frames accept a starred variant for the navigation arrow:

```
Refer to the \notionsframe*{definition of continuity}.
```

## 3. Numbering Systems

SCHOLIA provides two numbering systems for structured content. Each system offers both environment and inline command variants.

### a. Steps and Substeps

For procedural content with square-bracketed numbers. The substep counter resets automatically when a new step begins.

**Environment Synta**

```
\begin{step}[Preparation]
Gather all necessary materials.
\begin{substep}
Check equipment.
\end{substep}
\begin{substep}
Prepare workspace.
\end{substep}
\end{step}
```

**1 Preparation**

Gather all necessary materials. (a) Check equipment.

(b) Prepare workspace.

**Inline Syntax** For lighter markup in compact documents, inline commands are available but not currently exposed for steps. Use the environment form for steps.

### b. Questions and Subquestions

For assessment content with circled numbers. The subquestion counter (a, b, c, ...) resets automatically when a new question begins.

**Environment Synta**

```
\begin{question}
Solve $x^2 - 4 = 0$.
\begin{subquestion}
Factor the left side.
\end{subquestion}
\begin{subquestion}
Find all solutions.
\end{subquestion}
\end{question}
```

① Solve $x^2 - 4 = 0$. (a) Factor the left side.

(b) Find all solutions.

**Inline Syntax** For compact exercise sheets, use \q and \subq:

```
\q What is the capital of France?
\subq Name two rivers.
\subq Name the largest city.
```

Both syntaxes share the same counters and produce identical visual output.

### c. Counter Reset

| Counter | Reset Behavior |
|---|---|
| `scholia@stepnum` | Manual reset only |
| `scholia@substepnum` | Resets when `scholia@stepnum` increments |
| `scholia@qnum` | Resets when entering an exercise environment |
| `scholia@subqnum` | Resets when `scholia@qnum` increments |

## 4. Fiches (Worksheets)

Fiches are standalone worksheet layouts with styled banners and relative page numbering. They are available only in article mode (disabled in Beamer).

### a. Fiche Types

| Type Key | Color | Purpose |
|---|---|---|
| `notions` | Cyan | Concepts and theory |
| `exercices` | Orange | Exercise sheets |
| `corrections` | Green | Solutions |
| `autoevaluation` | Purple | Self-assessment |
| `methode` | Blue | Methods and procedures |
| `evaluation` | Red | Formal assessments |
| `sommaire` | Gray-brown | Chapter summaries |
| `aretenir` | Gold | Key points to memorize |
| `recherche` | Reddish-orange | Research activities |

**Note:** The `presentation` type exists for Beamer integration but is handled separately through the progression module in Beamer mode.

### b. Creating a Fiche

The `\newFiche` command initializes a new worksheet:

```
\newFiche{3e-func-N1}{notions}{Functions}{Images and Antecedents}
```

Arguments:

1. Reference identifier (for cross-referencing)
2. Type key (from table above)
3. Chapter title (displayed in gray)
4. Specific topic (displayed in bold)

This creates a banner with the appropriate icon, color, and titles, and sets up relative page numbering within the fiche.

### c. Full-Page Fiche Environments

For special emphasis, full-page framed environments wrap content in a distinctive border that spans the entire page. These are breakable across multiple pages.

**Summary Page** The `fichesummarypage` environment creates a light gray dotted frame for chapter summaries, competency inventories, or review sheets:

```
\begin{fichesummarypage}{ref}{Chapter}{Title}
Summary content here...
\end{fichesummarypage}
```

Arguments: reference identifier, chapter title, specific topic.

**Key Points Page** The `fichekeypointspage` environment creates a gold/bronze frame for essential concepts, formulas, or theorems to memorize:

```
\begin{fichekeypointspage}{ref}{Chapter}{Title}
Key points here...
\end{fichekeypointspage}
```

This environment suppresses header and footer rules for a cleaner presentation, maximizing focus on the content.

### d. Document Metadata

Fiches display author and year information in the footer. Define these in the preamble:

```
\renewcommand{\Auteur}{Author Name}
\renewcommand{\Annee}{2025--2026}
```

### e. Cross-Referencing

Reference fiches using the automatically created labels:

```
See \ficheRef{3e-func-N1} for the definition.
```

## 5. Exercises

SCHOLIA provides five exercise environment types. These types are *visual categories* that indicate the pedagogical context; the actual content within each environment is entirely flexible.

### a. Exercise Types

| Environment | Visual Indication |
|---|---|
| `exerciseguided` | Guided icon; suggests step-by-step assistance |
| `exercisesemiguided` | Semi-guided icon; partial hints expected |
| `exerciseautonomous` | Difficulty stars (1–3); independent work |
| `exercisemodeling` | Modeling icon; open-ended problem-solving |
| `exercisecorrection` | Correction styling; for displaying solutions |

### b. Exercise Environments

All exercise environments share a common structure: an optional title in brackets, followed by the exercise content. The `exerciseautonomous` environment additionally requires a mandatory difficulty level (1–3) displayed as stars.

**Standard exercise with titl**

```latex
\begin{exerciseguided}[Solving Quadratic Equations]
\exerciseStep{Identify coefficients $a$, $b$, $c$.}
\exerciseStep{Compute $\Delta = b^2 - 4ac$.}
\exerciseStep{Apply the quadratic formula.}
\end{exerciseguided}
```

**Exercise 1 – Solving Quadratic Equations** **Guided**

**a)** Identify coefficients $a$, $b$, $c$.

**b)** Compute $\Delta = b^2 - 4ac$.

**c)** Apply the quadratic formula.

**Autonomous exercise with difficulty leve**

```latex
\begin{exerciseautonomous}[Integration Practice]{2}
Compute $\int x^2 e^x \, dx$.
\end{exerciseautonomous}
```

**Exercise 2 – Integration Practice** ★★

Compute $\int x^2 e^x \, dx$.

### c. Two-Column Layout

The `exercisetwocolumns` environment displays content in two columns with a dotted separator, useful for compact exercise sheets:

```latex
\begin{exercisetwocolumns}
\begin{exerciseautonomous}{1}
Simplify $\frac{x^2 - 1}{x - 1}$.
\end{exerciseautonomous}

\begin{exerciseautonomous}{1}
Factor $x^2 + 5x + 6$.
\end{exerciseautonomous}
\end{exercisetwocolumns}
```

### d. Blanks and Hidden Solutions

SCHOLIA supports dual compilation: one version with answers hidden (for students) and one with answers visible (for teachers or answer keys). Blank commands insert placeholders that reveal their content only when solutions are enabled.

| | Command | Description |
|---|---|---|
| **Blank Commands** | `\solutionBlank{width}{answer}` | Underlined blank or answer |
| | `\solutionDotsBlank{n}{answer}` | Dotted blank (n dots) or answer |
| | `\fillBlank{width}` | Always show underlined blank |
| | `\blankLine{width}` | Simple underline |

Example usage:

```
The derivative is \solutionBlank{2cm}{$2x$}.
```

**Enabling Solutions** By default, solutions are hidden. Two methods enable solution display:

**Method 1: In the preamble**

```
\usepackage{scholia}
\solutionstrue
```

**Method 2: Via command line**

```
pdflatex "\def\ForceSolutions{}\input{document}"
```

This allows generating both student and teacher versions from the same source file.

### e. Exercise Commands

| Command | Description |
|---|---|
| `\exerciseStep[title]` | Numbered step (a, b, c, ...) |
| `\resetExerciseSteps` | Reset step counter to zero |
| `\highlight{text}` | Subtle gray background |
| `\questionStatement{text}` | Highlighted question in corrections |

### f. Counter Management

The exercise counter (`exonum`) increments automatically with each exercise environment. The step counter (`exoitem`) resets at each `\resetExerciseSteps` call. The question counter (`scholia@qnum`) resets automatically when entering any exercise environment.

## 6. Grading

The grading module provides tools for creating assessment rubrics and point allocation displays. These are intended for teacher documents (answer keys, grading sheets) rather than student-facing materials.

### a. Grading Box

A simple point allocation display:

```latex
\begin{gradingbox}[Point Distribution]
\gradingItem{Question 1}{2 pts}
\gradingItem{Question 2}{3 pts}
\gradingTotal{5 pts}
\end{gradingbox}
```

**Grading Scale – Point Distribution**

| | |
|---|---|
| **Question 1** | 2 pts |
| **Question 2** | 3 pts |
| **Total** | **5 pts** |

### b. Grading Table

For detailed rubrics with criteria descriptions:

```latex
\begin{gradingtable}{Assessment Rubric}
\gradingRow{Clarity}{Clear presentation}{2 pts}
\gradingRow{Accuracy}{Correct reasoning}{3 pts}
\gradingRowTotal{5 pts}
\end{gradingtable}
```

### c. Inline Point Indicators

Point values can be added inline within exercise text:

```latex
\begin{question}
Calculate $f(3)$. \gradingInline{2 pts}
\end{question}
```

### d. Automatic Point Tracking

The grading module includes counters for automatic point summation:

```latex
\resetGradingPoints
\gradingItemTracked{Q1}{Description}{2}
\gradingItemTracked{Q2}{Description}{3}
Total: \showGradingTotal
```

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## VII. Acknowledgments

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